Hussain Singapuri

GENERATIVE AI ENGINEER

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PROFESSIONAL SUMMARY

Generative AI Engineer with 3 years of professional software development experience at a US-based medical-device company, based in Pune. Builds **LLM applications** and **agentic AI** systems with tool calling and multi-agent orchestration (Python, asyncio, Claude API), **RAG** pipelines with hybrid BM25 + vector retrieval on pgvector, and **FastAPI** services, with evaluation harnesses and guardrails. Built an LLM inference engine from scratch (3.1x throughput over sequential decoding). Uses Claude Code daily. Claude Certified Developer.

CORE COMPETENCIES

Agentic AI & Multi-agent Orchestration | LLM Applications (Claude, Gemini APIs) | RAG, Vector Databases & Embeddings (pgvector) | Tool / Function Calling | Prompt Engineering & Evals | Python, asyncio & FastAPI | REST APIs & Docker | Transformer LLM Inference & Latency

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)
- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model
- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built **AI agents** for internal use
- Built and maintained full-stack web applications for the company
- Built several apps, including a CRM
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

multi-agent-research-crew

Open Source

Agentic AI: planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

site-to-agent

Open Source

RAG: crawler, hybrid BM25 + vector retrieval fused with RRF, cited answers, SSRF-hardened fetching and an eval harness for answer quality.

Next.js, TypeScript, pgvector

nanoserve Open Source

Transformer LLM inference engine built from scratch: paged KV cache, prefix caching, continuous batching with preemption, speculative decoding, OpenAI-compatible streaming API. 3.1x throughput over sequential decoding in benchmarks.

Python, PyTorch, FastAPI

ai-lead-receptionist Open Source

LLM agent for home-service businesses: missed-call text-back, SMS and web chat that qualifies leads and books jobs, with an owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

lead-intel-engine Open Source

Discovers, enriches and scores local-business leads with a robots-aware, rate-limited crawler and transparent scoring.

Python, FastAPI, SQLite

contractor-ai-ops-dashboard Open Source

Multi-tenant analytics with signed event ingestion, funnel metrics, anomaly alerts and number-checked AI weekly summaries.

Next.js, TypeScript, Recharts, Postgres RLS

PageForge Portfolio project

Guardrails for LLM output: AI drafts only from approved facts and the account's public pages, a deterministic validator flags unsupported numbers, and a reviewer signs off claim by claim.

Next.js 16, React 19, Sanity, Supabase, Gemini

TripSync Project

AI business-travel planner: enter trip constraints, get three AI-generated packages, refine via chat, export to calendar.

Flutter, Supabase, Gemini structured outputs, GitHub Actions

More projects

Personal portfolio with full-stack backend, Orbital Surplus, Nightjar & Co., Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club, Safey Inventory

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

AI / LLM: AI agents, multi-agent orchestration, tool use / function calling, RAG (hybrid BM25 + vector, RRF), evals, prompt-injection defence, LLM inference serving, Claude API, Gemini API, Claude Code and AI coding agents

Languages: Python, TypeScript, JavaScript, SQL, Dart, HTML, CSS

Back end: FastAPI, asyncio, REST APIs, Node.js, webhooks (HMAC signing), Stripe Checkout and refunds, auth (scrypt, sessions, throttling), Resend

Data: pgvector, PostgreSQL, Neon, Supabase (RLS, Edge Functions), PGlite, SQLite, Sanity CMS

DevOps: Docker, GitHub Actions, Vercel

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Front end: Next.js, React 19, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO